Technical specification template: Power Transformers

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for power transformers for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 60076 series (power transformers)
- IEC 60076-7 (loading guide, mineral-oil-immersed)
- IEC 60076-11 (dry-type), if dry-type is selected
- IEC 60076-5 (ability to withstand short circuit)
- Local utility and authority requirements [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit	NOT CALCULATED

	study pending utility, transformer and generator data]	

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Quantity (total)	2	PRELIMINARY_ASSUMPTION
Rated power (each)	3,150 kVA	UNVERIFIED_PLACEHOLDER. Loading limits are applied to the derated rating
Short-circuit impedance	6 %	UNVERIFIED_PLACEHOLDER. Guaranteed value and tolerance from the manufacturer
Primary / secondary voltage	13.8 kV / 400 V	PRELIMINARY_ASSUMPTION
Type and cooling	[Dry-type or liquid-immersed; cooling class: project to select]	Project decision
Vector group, tap changer	[Project to select]	Project decision
Derating (ambient, altitude, installation)	1.0	UNVERIFIED_PLACEHOLDER. 1.00 is a placeholder. Manufacturer's loading guide governs

5. Technical requirements

5.1 The transformer shall carry its rated power continuously at the site conditions in the service-conditions table without exceeding the temperature-rise limits of the applicable standard.

5.2 The manufacturer shall state the continuous rating at the site conditions and any derating for ambient temperature, altitude and enclosure.

5.3 The manufacturer shall not rely on short-duration emergency overload to meet the continuous loading requirement.

5.4 The short-circuit impedance, tolerance, no-load and load losses shall be guaranteed values supplied in the tender and confirmed by test.

5.5 The transformer shall withstand the thermal and dynamic effects of the specified through-fault level for the specified duration; evidence by test or calculation per the applicable standard.

5.6 Losses evaluation basis: [project to define].

5.7 Noise level, enclosure protection rating, temperature monitoring and protection accessories: [project to define].

5.8 Number of transformers per power path: 1; each path is to carry the whole facility load after loss of the other path.

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
equipment.transformer_rating	Transformer rating	Transformer data sheet	UNVERIFIED_PLACEHOLDER
equipment.transformer_impedance	Transformer impedance	Transformer data sheet	UNVERIFIED_PLACEHOLDER
losses.distribution_loss_fraction	Distribution losses	Transformer loss report and busduct and cable loss data	UNVERIFIED_PLACEHOLDER
derating.transformer_ambient_factor	Transformer ambient derating factor	Manufacturer derating curve for ambient conditions (transformer)	UNVERIFIED_PLACEHOLDER
derating.transformer_altitude_factor	Transformer altitude derating factor	Manufacturer derating curve for altitude conditions (transformer)	UNVERIFIED_PLACEHOLDER
derating.transformer_installation_factor	Transformer installation derating factor	Manufacturer derating curve for installation conditions (transformer)	UNVERIFIED_PLACEHOLDER
derating.transformer_contingency_factor	Transformer contingency derating factor	Transformer data sheet	UNVERIFIED_PLACEHOLDER
short_circuit.utility_fault_level_mva	Utility initial short-circuit power at the MV connection	Utility written statement of maximum and minimum fault levels and X/R at the point of connection	MISSING
short_circuit.transformer_x_r_ratio	Transformer X/R ratio	Transformer test report (short-circuit impedance and load losses)	MISSING

7. Inspection and quality requirements

7.1 Review of design calculations, drawings and the inspection and test plan before manufacture.

7.2 Inspection of materials certificates and core and winding processing at hold points defined in the ITP [project to define].

7.3 Dimensional and nameplate check before shipment.

7.4 Routine tests on every unit as required by the applicable standard and this specification.

7.5 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.6 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.7 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Routine tests per the standard: ratio and vector group, winding resistance, no-load loss and current, load loss and impedance voltage, applied voltage, induced voltage, insulation resistance.

8.2 Temperature-rise test or evidence on a duplicate unit [project to define].

8.3 Witnessing: [owner / independent inspector to attend]. Results recorded against the guaranteed values.

8.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Visual and mechanical inspection, torque checks, insulation resistance, ratio and polarity.

9.2 Protection and temperature-alarm function tests.

9.3 Energisation sequence and thermography under load [project to define].

9.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.